

Procedure: Make sure to carefully follow all the prompts and questions. Refer back to your study and book as you need to.

1. The model to the right illustrates how nature recycles what natural resources?

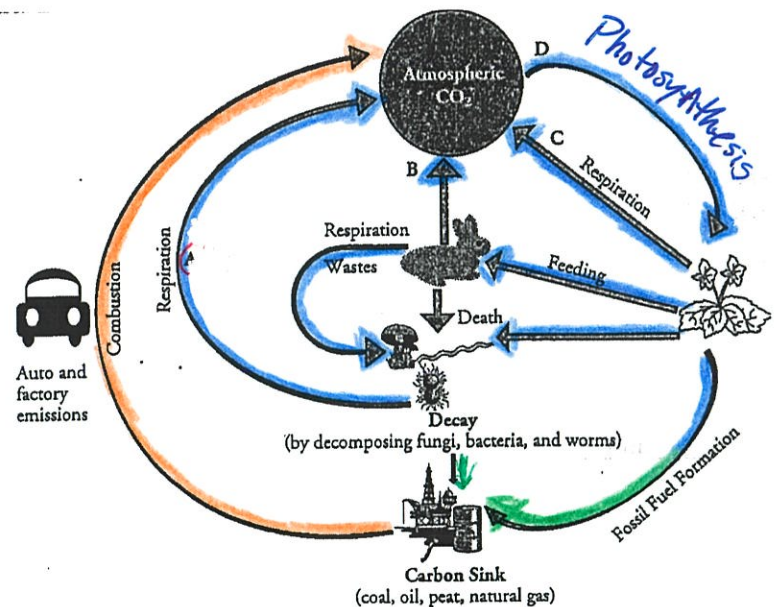
Carbon

2. Name **2 ways** that carbon (usually in the form of CO_2) enters the atmosphere.

- Respiration
- Combustion

3. **Process D** on the diagram uses CO_2 from the atmosphere.

- LABEL D on the diagram in Model 2 with the name of the process used.
- What organisms carry out the process identified in PART A? Producers / Plants



4. Wastes and dead organisms must be broken down in order for their components to be used again.

- What organisms in the cycle carry out this process?
decomposing fungi, bacteria, worms
- What would happen if decomposition did not occur?
Animals & plants would not decay, returning carbon to the soil and atmospheric CO_2 would decrease

5. Using **textbook pg. 83, COLOR** the arrows/ process according to the key below:

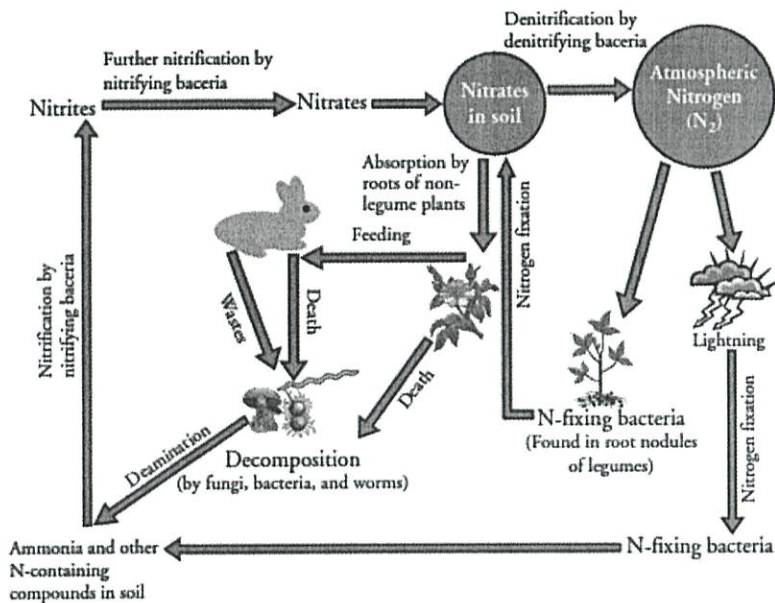
- Biological = Blue
- Physical/ Chemical = RED
- Geologic = Green
- Human activities = ORANGE

6. Not all dead organisms are acted on by decomposers. Instead of being immediately recycled, the carbon from some organisms is kept in a type of long-term storage, or carbon sink. Using model 2 to answer the following questions.

- List **4 materials** that contain this stored carbon Coal, oil, peat, natural gas
- What is the term we use to describe these materials? Fossil fuels (Carbon Sink)
- Which **type of process** is responsible for this storage? (use question 5 for processes)
Geological Process

7. Complete the assigned Interactivity: Biogeochemical Cycles

READ THIS!!! Nitrification is a process by which specific bacteria convert different forms of N-containing compounds (like ammonia, NH_3) in the soil to nitrites (NO_2) and nitrates (NO_3). **This process is important since the only forms of nitrogen that are usable by plants to build their proteins and pass it to animals are the nitrates.**



1. This model illustrates how nature recycles what natural resource? **Nitrogen**
2. In what ways is N_2 gas removed from the atmosphere? **Lightning & Nitrification**
3. What organisms are essential in recycling animal wastes and dead organisms? What process is this? **Decomposers (fungi, bacteria, worms)** **Decomposition**
4. What is the only form of nitrogen that plants can take in and use? **Nitrates**

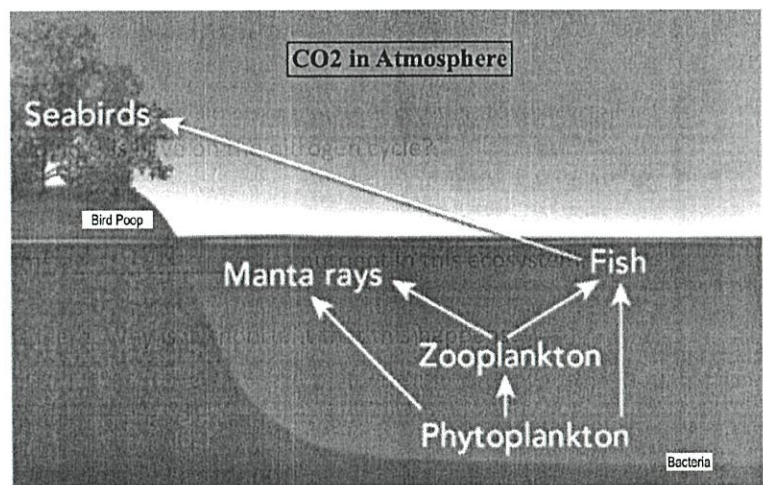
5. Nitrogen contained in plants gets turned into what when herbivores feed on them? **Protein**
6. Humans can add nitrogen to the biosphere in the form of what? Where would this be added to the nitrogen cycle? **Fertilizer in the form of Nitrates. This would be added at "Nitrates in Soil"**
7. Nitrification creates usable / unusable (circle one) and Denitrification creates usable / unusable (circle one).
8. During the **denitrifying** process, what happens to nitrates? **Nitrates are turned back into unusable Atmospheric N_2**
9. If the number of nitrifying bacteria decreased, what effect would this have on the nitrogen cycle? **Ammonia & other N-containing compounds would build up in the soil. Nitrates would hardly be produced & plant would decrease as well as animals.**
10. If the above situation happened, nitrogen would then become a limiting nutrient in this ecosystem.
11. Some Usuable nitrogen (nitrates) are returned to the atmosphere. Why is it important that this happens? **Then ecosystems/biomes without many plants would still be able cycle nitrogen and not have it be a limiting nutrient**

Applying Biogeochemical Processes to a food web

In the diagram below, there is a basic food web in a coastal ocean area.

→ To the diagram, Add in the following terms where they apply.
You will need to add arrows to this diagram

Photosynthesis
Respiration
Evaporation
Transpiration
Dissolves
Decomposition
Feeding
Waste
Runoff

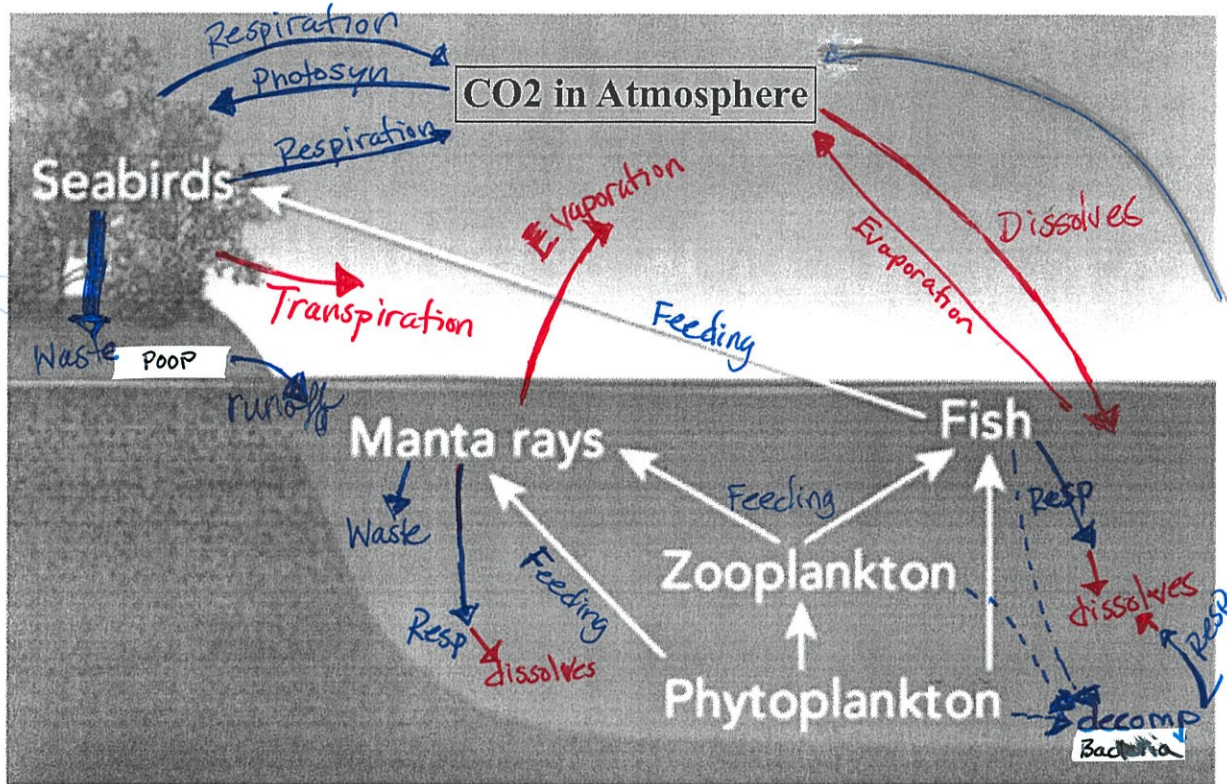


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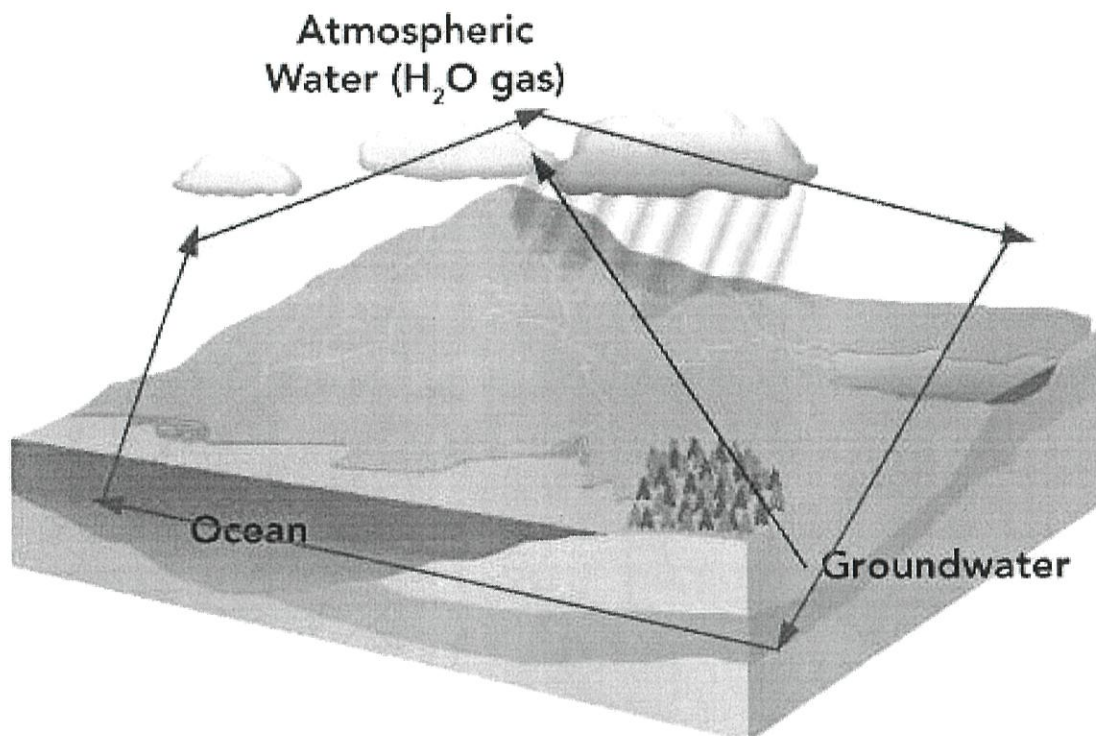
Photosynthesis, Cellular respiration, Evaporation, Transpiration, Dissolves, Decomposition, Feeding, Waste, runoff



* you didn't need to use all these arrows.
Just giving you possible options!

Biogeochemical Cycles

Screen 1 #1: Answers will vary. Sample answer:



Screen 3 #2: photosynthesis; cellular respiration

Screen 4 #3: Removes #4: Adds #5: Removes #6: Adds #7: Adds

Screen 6 #1: fertilizer. #2: bacteria. #3: atmosphere.